

Akshat Sharma

B.Tech Computer Science Student • Full-Stack Systems & Tooling

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Professional Summary

B.Tech CSE student specializing in full-stack web architectures, event-driven backend performance engineering, and local desktop automation developer tools.

Technical Skills

Languages:	JavaScript (ES6+), TypeScript, Python, HTML5, CSS3
Backend & Databases:	Node.js, Express.js, WebSockets (Socket.IO), Redis Caching, MongoDB, Mongoose, REST APIs
Frontend & UI:	React.js, Next.js, Tailwind CSS, Figma (UI/UX Component Design)
AI Tooling & Infrastructure:	LLM API Integration, WebContainers, Telegram Bot API, Git, GitHub, Artillery Testing

Projects

FLUX — Collaborative Workspace Architecture	Live Workspace — Repository — [Video Walkthrough]
<i>Full-Stack Core Infrastructure Project</i>	

- Designed and engineered backend authentication layers, integrating JWT tokens with a server-side Redis validation cache to handle explicit session revocation and token blacklisting.
- Implemented a 300ms debounced state-synchronization layer to decouple volatile incoming WebSocket data streams from synchronous React component re-renders, preventing remote cursor focus drift.
- Benchmarked system scalability using Artillery load-testing suites to simulate 300 concurrent connections, validating an empirical p95 message latency threshold of 1.1ms.

ShadowClaw — Context-Aware CLI Developer Tooling	[Demo Video]
<i>TypeScript Engineering Framework</i>	

- Built a modular TypeScript command-line utility featuring separate execution runtimes for immediate context retrieval, directory structural tree parsing, and structured code block output generation.
- Configured automated workspace context ingestion by programmatically parsing local structured Markdown directories, mapping nested index routes into structured LLM string templates.

Friday — System-Tray Desktop Automation Utility	[Demo Video]
<i>Native Background Service</i>	

- Developed a native desktop background loop using local speech-to-text processing layers to translate incoming voice commands into local shell command execution and child_process execution channels.
- Implemented an asynchronous, native node file-system monitor (fs.watch) that tracks workspace directories to trigger underlying ShadowClaw sub-processes, delivering sub-second local code template output.

Education

Bachelor of Technology — Computer Science & Engineering	Sep 2023 – Jun 2027 (Expected)
Malwa Institute of Technology, Indore, Madhya Pradesh	

Achievements & Activities

- Built and presented rapid working software prototypes under strict operational constraints at national-level hackathons, including the BGI Hackathon 2026 (Cybersecurity track) and Tech-o-tsav 2024.